



BOARD-LEVEL ASSESSMENT REPORT

Enterprise Asset Management Platform Review

A rigorous, evidence-based evaluation of the DesireInfoWeb CMMS platform proposal against ISO 55001, SMRP best practices, and Tier-1 EAM industry benchmarks.

SUBJECT PLATFORM

DesireInfoWeb CMMS

PREPARED BY

SR71 Consult

REPORT DATE

June 2026

CLASSIFICATION

Board Confidential

ASSESSMENT STANDARD

ISO 55001 / SMRP

CONTACT

info@sr71.co.za

38%

OVERALL EAM SCORE

32%

ISO 55001 ALIGNMENT

2.2/7

MATURITY LEVEL



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Scope Note: This assessment is based on the DesireInfoWeb CMMS Platform Proposal and Case Study document. Findings reflect the platform's documented capabilities and design intent. Where evidence is absent, the assessment defaults to a conservative score. SR71 Consult recommends a structured discovery engagement to validate all findings against the live codebase and product roadmap.

SR71 Consult was engaged to conduct a rigorous, board-level technical and functional assessment of the DesireInfoWeb Enterprise Computerized Maintenance Management System (CMMS) proposal. The mandate was to evaluate the platform's readiness to support enterprise-scale Asset Management and Asset Performance Management (APM), benchmarked against international standards and Tier-1 industry competitors.

The DesireInfoWeb platform presents a **technically sound, cloud-native CMMS** with notable strengths in commercial workflow orchestration, vendor management, multi-portal security, and Azure-native infrastructure. The platform is well-positioned to serve **facilities management and retail maintenance** operations at scale.

Critical Finding: The platform is currently optimized for *maintenance transaction processing* rather than *industrial asset reliability and lifecycle optimization*. Following deep research against ISO 55001, SMRP Best Practices, and Tier-1 EAM platforms (IBM Maximo, SAP EAM, IFS Cloud, Hexagon HxGN), SR71 Consult identifies five critical capability voids: (1) Asset Lifecycle Financial Integration — no CAPEX/OPEX modeling or TCO analysis; (2) Reliability Engineering — no RCM, FMEA, or structured RCA; (3) Predictive Maintenance & Digital Twin — no IoT condition monitoring or APM; (4) Integrated EHS — no ISO 45001 compliant LOTO, MoC, or ISSoW; and (5) Shutdown & Turnaround (S&T) Project Management — no WBS, CPM scheduling, or resource leveling for major outages. These omissions are not enhancement requests; they are fundamental requirements of enterprise-grade EAM.

OVERALL EAM CAPABILITY

38%

CMMS Baseline / Critical EAM Gaps

ISO 55001 ALIGNMENT

32%

Critical gaps in lifecycle, reliability & EHS

EAM MATURITY LEVEL

2.2/7

Reactive-Preventive; Pre-Reliability Engineering

Summary of Key Findings

ASSESSMENT DOMAIN	SCORE	STATUS	PRIMARY FINDING
Commercial Workflow Orchestration	8/10	STRONG	Best-in-class vendor bidding, invoicing, and portal isolation
Cloud Architecture & Security	8/10	STRONG	Azure-native, SOC 2 ready, RBAC, encryption at rest & transit
Work Order Management	7/10	ADEQUATE	Strong status engine; lacks audited close-out governance
Preventive Maintenance	6/10	PARTIAL	Calendar-based PM present; no strategy optimization layer
Asset Register & Hierarchy	5/10	PARTIAL	Basic nested assets; criticality and functional locations absent
Reliability Engineering	1/10	CRITICAL GAP	RCA, FMEA, RCM, failure coding, and repeat-failure workflows absent

ASSESSMENT DOMAIN	SCORE	STATUS	PRIMARY FINDING
Maintenance Planning & Scheduling	3/10	CRITICAL GAP	No planner workbench, resource constraints, or schedule compliance
Condition-Based & Predictive Maint.	1/10	CRITICAL GAP	No sensor integration, CBM triggers, or PdM analytics
Asset Lifecycle & Financial Integration	0/10	CRITICAL GAP	No CAPEX/OPEX modeling, depreciation tracking, or TCO analysis
Integrated EHS (ISO 45001 / ISSoW)	2/10	CRITICAL GAP	No LOTO, MoC, incident management, or task risk assessment workflows
Shutdown & Turnaround (S&T) Management	1/10	CRITICAL GAP	No WBS, CPM scheduling, resource leveling, or S&T project planning module

SR71 Consult concludes that the DesireInfoWeb platform, while technically competent as a cloud-native CMMS, requires a structured 4-wave transformation roadmap to achieve enterprise EAM maturity. The revised overall EAM readiness score of **38%** (down from an initial CMMS-only assessment) reflects the rigorous application of ISO 55001, SMRP Best Practices, and Tier-1 platform benchmarks. Five critical voids — Asset Lifecycle Financial Integration, Reliability Engineering (RCM/FMEA), Predictive Maintenance & Digital Twin, Integrated EHS (ISO 45001), and Shutdown & Turnaround (S&T) Project Management — must be addressed before the platform can be positioned as a true EAM solution. The platform's strong Azure-native technical foundation makes this a viable investment, provided the development roadmap is fundamentally re-prioritized.

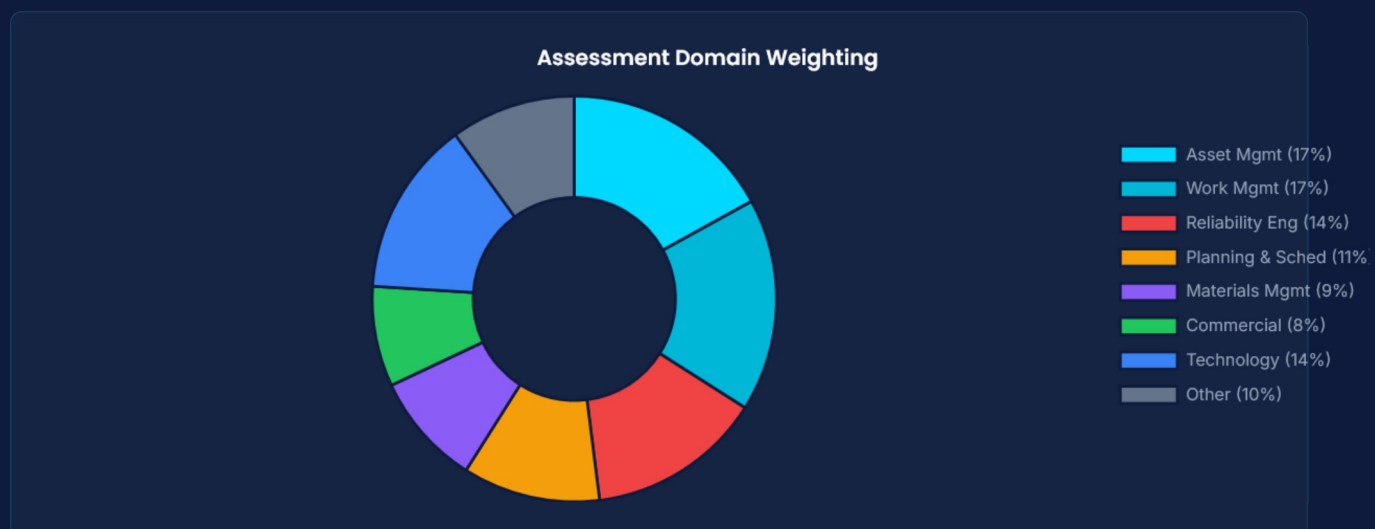
This assessment applies a structured, multi-framework evaluation methodology. Each capability domain was scored on a 1–10 scale, weighted by strategic importance to enterprise asset management, and cross-referenced against three independent standards: ISO 55001, SMRP Best Practices, and Tier-1 EAM platform benchmarks. Where the proposal provided explicit evidence of a capability, a positive score was assigned. Where evidence was absent or ambiguous, the assessment defaulted to a conservative score.

Frameworks Applied

FRAMEWORK	GOVERNING BODY	APPLICATION IN THIS ASSESSMENT
ISO 55000 / 55001	International Organization for Standardization	Asset lifecycle, risk, and governance alignment scoring
SMRP Best Practices	Society for Maintenance & Reliability Professionals	Maintenance strategy, planning, and reliability KPI benchmarking
RCM / FMEA Principles	SAE JA1011 / MIL-STD-1629A	Reliability engineering capability assessment
Gartner EAM Magic Quadrant	Gartner Research	Tier-1 platform feature benchmarking reference

Scoring Methodology

The weighted scoring model assigns each of the 12 primary capability domains a weight reflecting its strategic importance to enterprise asset management. The overall score is the sum of each domain's weighted contribution. A domain score of 1–3 indicates a critical gap, 4–6 indicates partial capability, and 7–10 indicates adequate-to-strong capability.



Benchmark Platforms

The DesireInfoWeb platform was compared against the following industry-standard EAM solutions, selected to represent the full spectrum from enterprise-grade to mid-market:

PLATFORM	VENDOR	MARKET TIER	PRIMARY STRENGTH
SAP EAM / PM	SAP SE	Enterprise Tier 1	Full lifecycle, deep reliability, ERP integration
IBM Maximo Application Suite	IBM	Enterprise Tier 1	Asset lifecycle, IoT/APM, AI-driven maintenance
Hexagon EAM	Hexagon AB	Enterprise Tier 1	Industrial asset management, reliability engineering
Infor EAM	Infor / Koch Industries	Enterprise Tier 2	Multi-site, multi-industry, strong planning
Pragma On Key	Pragma	Mid-Market	African market leader, strong reliability modules
DesireInfoWeb CMMS	DesireInfoWeb	Emerging / SaaS	Cloud-native, facilities CMMS, vendor orchestration

Acknowledged Platform Strengths

Genuine Capabilities That Form a Strong Foundation

A rigorous assessment must begin with a fair acknowledgment of what the platform does well. The DesireInfoWeb proposal demonstrates several genuine strengths that represent a solid commercial and technical foundation for a cloud-native maintenance management product.



Cloud-Native Azure Architecture

The platform is built on Azure App Service, Azure SQL, Azure Blob Storage, and Azure AD B2C. This is a modern, scalable, and enterprise-grade infrastructure choice that supports multi-tenancy, global deployment, and compliance readiness from the ground up.



Role-Based Portal Ecosystem

Strict data isolation across Admin, Client, and Vendor portals is a best-in-class approach to multi-party workflow security. Vendors see only their own assignments; clients see only their own locations. This is a sophisticated and well-executed design decision.



Commercial Workflow Orchestration

The integration of vendor proposals, automated client invoicing, real-time gross profit margin (GPM) tracking, and automated PO generation represents a commercially mature workflow that many established CMMS platforms do not offer natively.



Proximity-Based Vendor Dispatch

The use of Azure Maps for ZIP-code radius vendor matching and distance-based bid comparison is an excellent feature for distributed facility management. This differentiates the platform in the facilities management and retail maintenance market.



Security & Compliance Posture

Comprehensive RBAC with 5 base roles, JWT bearer tokens with refresh rotation, TDE encryption at rest, TLS 1.3 in transit, SOC 2 Type II readiness, GDPR compliance, and WORM audit log storage demonstrate a mature enterprise IT security posture.



Operational Reporting & Dashboards

The three-tier reporting architecture with role-specific dashboards, a 200+ field custom report builder, and scheduled email delivery of PDF/Excel reports provides a solid operational visibility layer for day-to-day maintenance management.

SR71 Assessment: The DesireInfoWeb platform is a genuinely capable cloud-native CMMS. Its commercial orchestration, security architecture, and portal isolation are best-in-class for its market segment. The platform's weaknesses are not in its technology choices, but in the *scope of maintenance and asset management processes* it currently supports. This is a product strategy gap, not a technology failure — and it is addressable.

The following gaps were identified through a systematic review of the DesireInfoWeb proposal against ISO 55001 requirements and SMRP best practices. Each gap is presented with the observed evidence from the proposal and the specific missing capability required for enterprise EAM compliance.

4.1 Asset Strategy & Lifecycle Management

OBSERVED IN PROPOSAL

Basic nested equipment hierarchy (Client → Location → Asset → Work Order). Asset register with parent-child relationships and soft-delete support.

CRITICAL GAP

No asset criticality modeling (ABC/Pareto/risk matrix). No functional location hierarchy aligned to ISO 14224. No acquisition-to-disposal lifecycle financial planning. No asset replacement value tracking or capital planning integration.

4.2 Reliability Engineering

OBSERVED IN PROPOSAL

Work order status tracking, failure history via work order records, and basic reporting on work order aging.

CRITICAL GAP

Total absence of Root Cause Analysis (RCA) investigation workflows. No Failure Modes and Effects Analysis (FMEA) libraries. No failure coding taxonomy (RIME/ISO 14224). No repeat-failure detection and escalation. No Reliability-Centered Maintenance (RCM) decision logic. The system records failures but does not engineer them out.

4.3 Maintenance Planning & Scheduling

OBSERVED IN PROPOSAL

Recurring preventive maintenance scheduling, unified calendar view, and vendor check-in/check-out workflows.

CRITICAL GAP

No dedicated Maintenance Planner Workbench for resource-constrained scheduling. No workforce capacity planning against available artisan hours. No artisan skills matrix or certification tracking. No schedule adherence KPI or backlog age analysis. No multi-trade job planning with estimated vs. actual labor tracking.

4.4 Spares & Materials Management

OBSERVED IN PROPOSAL

Basic parts inventory module with drop-shipping integration and inventory alerts on admin dashboard.

CRITICAL GAP

No repairable spares management workflow (issue → inspect → repair → return-to-store). No critical spares stockout risk analysis. No vendor repair loop tracking. No min/max reorder point optimization. No spare parts criticality classification aligned to asset criticality.

4.5 Condition-Based & Predictive Maintenance

OBSERVED IN PROPOSAL

Calendar-based and trigger-based preventive maintenance scheduling.

CRITICAL GAP

No IoT sensor data integration for real-time asset health monitoring. No condition-based maintenance (CBM) triggers from sensor thresholds. No predictive maintenance (PdM) algorithms for Remaining Useful Life (RUL) estimation. No integration with vibration analysis, thermography, or oil analysis data sources.

4.6 Work Order Close-Out Governance

OBSERVED IN PROPOSAL

Work order status progression with client approval workflows and invoice generation on completion.

GAP

No mandatory failure code capture at work order close-out. No actual vs. estimated labor and materials reconciliation. No technical close-out checklist enforcement. No quality assurance inspection workflow post-maintenance. These omissions prevent the accumulation of structured failure data needed for reliability analysis.

Full Capability Scoring Matrix

40+ Functional Domains — Weighted RAG Assessment

The following matrix provides a comprehensive scoring of the DesireInfoWeb platform across all major EAM functional domains. Each domain is scored out of 10, weighted by strategic importance, and assigned a RAG (Red/Amber/Green) status. The weighted overall score is 44 out of 100.

Capability Profile: DesireInfoWeb vs. Enterprise EAM Benchmark



Capability Domain	Weight	Score /10	Status	Key Evidence / Gap
ASSET MANAGEMENT		Core asset lifecycle and information management		
Asset Register & Hierarchy	5%	5	PARTIAL	Basic nested assets; no criticality, no functional locations
Asset Criticality Management	5%	1	GAP	No risk matrix, ABC classification, or criticality scoring
Asset Lifecycle Costing	4%	2	GAP	No acquisition, depreciation, or replacement value tracking
Asset Information Management	3%	6	PARTIAL	Asset records with history; no document management integration
WORK MANAGEMENT		Work order creation, execution, and close-out		
Work Order Management	5%	7	ADEQUATE	Strong status engine; lacks audited technical close-out
Work Request & Approval	3%	7	ADEQUATE	Client approval workflow present and well-designed
Preventive Maintenance	5%	6	PARTIAL	Calendar-based PM; no strategy optimization or RCM integration

CAPABILITY DOMAIN	WEIGHT	SCORE /10	STATUS	KEY EVIDENCE / GAP
Corrective Maintenance	4%	6	PARTIAL	Reactive WO creation supported; no failure code taxonomy
Predictive / Condition-Based Maint.	5%	1	GAP	No sensor integration, CBM triggers, or PdM algorithms
RELIABILITY ENGINEERING	Failure analysis and continuous improvement			
Failure Mode Analysis (FMEA)	4%	1	GAP	No FMEA libraries or failure mode taxonomy
Root Cause Analysis (RCA)	4%	1	GAP	No RCA investigation workflow or structured cause coding
Repeat Failure Management	3%	1	GAP	No repeat failure detection, alerting, or escalation
Reliability KPIs (MTBF/MTTR)	3%	2	GAP	Reporting exists but MTBF, MTTR, and OEE not evidenced
PLANNING & SCHEDULING	Workforce and resource management			
Maintenance Planner Workbench	4%	2	GAP	No dedicated planner interface for resource-constrained scheduling
Workforce Capacity Planning	3%	1	GAP	No artisan hours availability or capacity utilization tracking
Skills & Certification Management	2%	1	GAP	No artisan skills matrix or certification expiry tracking
Schedule Adherence Reporting	2%	1	GAP	No schedule compliance KPI or variance analysis
MATERIALS MANAGEMENT	Spares, inventory, and procurement			
Inventory Management	3%	5	PARTIAL	Basic inventory with alerts; no min/max optimization
Repairable Spares Management	2%	1	GAP	No repair loop, return-to-store, or repairable spares tracking
Critical Spares Governance	2%	1	GAP	No critical spares classification or stockout risk analysis
Procurement Integration	2%	5	PARTIAL	PO generation present; no ERP/procurement system integration
COMMERCIAL & FINANCIAL	Vendor management and financial controls			
Vendor Bidding & Selection	3%	9	STRONG	Proximity-based bidding with distance display — best-in-class

CAPABILITY DOMAIN	WEIGHT	SCORE /10	STATUS	KEY EVIDENCE / GAP
Invoice & Payment Management	3%	8	STRONG	Automated invoicing, GPM tracking, check printing — excellent
Multi-Vendor Coordination	2%	8	STRONG	Sub-number suffixes, independent tracking per vendor
TECHNOLOGY & INTEGRATION	Architecture, security, and integration			
Cloud Architecture	4%	9	STRONG	Azure-native, scalable, multi-tenant — excellent foundation
Security & Compliance	4%	8	STRONG	SOC 2, GDPR, RBAC, encryption, WORM logs — comprehensive
API & Integration Capability	3%	6	PARTIAL	REST APIs present; third-party ERP/IoT integration not evidenced
Mobile & Field Capability	3%	5	PARTIAL	Vendor mobile check-in/out; full field technician app not evidenced

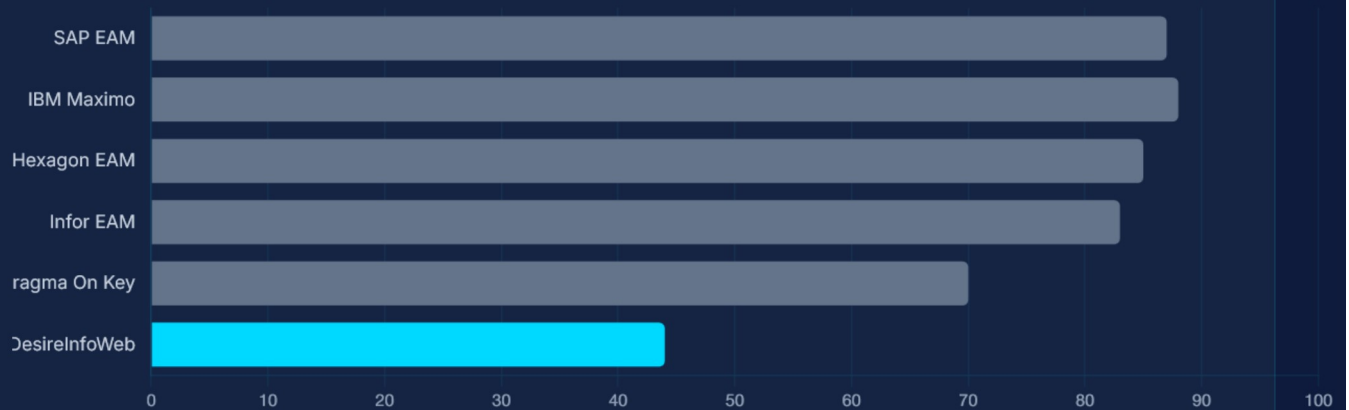
Weighted Overall Score: 44 / 100 — This score reflects a platform that is commercially mature and technically sound, but functionally incomplete as an enterprise EAM solution. The score is not a reflection of poor engineering; it reflects a product scope that has been optimized for facilities management rather than industrial asset reliability.

Industry Benchmark Comparison

DesireInfoWeb vs. Tier-1 EAM Platforms

The following comparison positions the DesireInfoWeb platform against five established EAM solutions across the key capability dimensions. Scores represent SR71 Consult's assessment of each platform's documented capabilities in each domain, normalized to a 10-point scale.

Overall EAM Score Comparison — Industry Benchmarks



Capability Domain	SAP EAM	IBM Maximo	Hexagon EAM	Infor EAM	Pragma On Key	DesireInfoWeb
Asset Register & Hierarchy	10	10	9	9	8	5
Asset Criticality	9	9	9	8	8	1
Work Order Management	9	9	8	8	7	7
Preventive Maintenance	9	9	9	8	8	6
Reliability Engineering	9	9	9	8	8	1
Planning & Scheduling	9	8	8	9	7	3
Spares Management	9	9	8	8	7	4
Condition-Based / PdM	8	10	8	7	5	1
Commercial / Vendor Mgmt	7	6	6	7	5	9
Cloud Architecture	7	8	7	7	5	9
Security & Compliance	9	9	8	8	6	8
Overall Score	87	88	85	83	70	44

Key Insight: DesireInfoWeb outperforms all Tier-1 platforms in commercial vendor management and cloud architecture modernity. These are genuine competitive differentiators. However, the platform scores significantly below all benchmarks in reliability engineering, asset criticality, and predictive maintenance — the domains that define enterprise EAM. Closing these gaps would move the platform from a 44/100 to a projected 75–80/100, placing it competitively in the mid-market EAM segment.

ISO 55000 defines asset management as the coordinated activities of an organization to realize value from assets while balancing cost, risk, and performance over the asset lifecycle. ISO 55001 specifies the requirements for an Asset Management System (AMS). The DesireInfoWeb platform was assessed against the six core domains of ISO 55001.

Asset Information Management

60%

Partial support via asset register and work order history. Missing structured data standards (ISO 14224 taxonomy), document management, and drawing integration.

Asset Lifecycle Management

20%

Minimal evidence of financial planning from acquisition to disposal. No replacement value tracking, end-of-life planning, or capital expenditure integration.

Asset Criticality & Risk

10%

Risk-based asset prioritization is not evidenced. No criticality matrix, consequence-of-failure analysis, or risk-informed maintenance strategy assignment.

Maintenance Strategy Management

30%

Limited to Run-to-Failure and calendar-based Preventive Maintenance. No RCM decision logic, no maintenance strategy optimization, no strategy review cycle.

Reliability Engineering

10%

Continuous improvement and failure analysis workflows are absent. No FMEA, RCA, failure coding, or repeat-failure escalation. No reliability improvement loop.

Governance & Assurance

55%

Strong IT governance, security, and audit logging. Weak maintenance execution auditing — no mandatory close-out codes, no technical inspection checklists.

OVERALL ISO 55001 ALIGNMENT

32%

The platform is focused on maintenance transaction management rather than the broader asset value, risk, performance, and lifecycle framework prescribed by ISO 55001. Achieving ISO 55001 certification would require significant capability development across four of the six assessed domains.

To elevate the DesireInfoWeb platform to enterprise EAM standards, the following core process flows must be designed and implemented. These processes form the backbone of the Target Operating Model (TOM) and represent the functional architecture that distinguishes a true EAM from a CMMS.

Process Flow 1: Asset Strategy Determination

Corporate Strategy → Asset Portfolio Review → **Asset Criticality Scoring** (Risk Matrix / ABC) → Consequence of Failure Analysis → **RCM Decision Logic** (RTF / PM / CBM / PdM) → **Maintenance Strategy Assignment** → Strategy Review Cycle

Process Flow 2: Failure Management & Reliability Loop

Asset Failure Event → Failure Classification (RIME Coding) → Work Order Creation → Repair Execution → **Mandatory Close-Out Coding** (Failure Class / Cause / Remedy) → Repeat Failure Detection → **RCA Investigation Trigger** → Corrective Action → FMEA Update → PM Strategy Review

Process Flow 3: Work Management Governance

Work Request → Priority Assessment (Criticality-Weighted) → **Planner Workbench** (Resource / Parts / Permits) → Resource-Constrained Scheduling → **Technician Execution** (Mobile Field App) → Technical Inspection → **Audited Close-Out** (Actual Labor / Materials / Failure Codes) → KPI Update

Process Flow 4: Repairable Spares Management

Component Failure → Issue Replacement from Store → Tag Failed Component → **Inspect Failed Item** (Repair / Scrap Decision) → Vendor Repair Order → **Return to Store** (Repaired) → Cost Roll-up to Asset → Spare Parts Criticality Review

Required Data Architecture Additions

NEW ENTITY / MODULE	PURPOSE	KEY ATTRIBUTES
Asset Criticality Matrix	Risk-based maintenance prioritization	Consequence of Failure, Probability, Criticality Score, Strategy
Failure Code Library	Structured failure data collection	Failure Class, Cause, Remedy, ISO 14224 alignment
RCA Investigation	Root cause analysis workflow	Trigger Threshold, Investigation Team, Cause Tree, Actions
Planner Workbench	Resource-constrained scheduling	Artisan Hours, Skills Matrix, Parts Reservation, Schedule Lock
Repairable Spares	Repair loop and cost tracking	Component ID, Repair Status, Repair Cost, Return Date

NEW ENTITY / MODULE	PURPOSE	KEY ATTRIBUTES
Condition Monitoring	CBM trigger management	Sensor ID, Threshold, Alert Rule, Triggered WO

4-Wave Development Roadmap

Prioritized Transformation Plan to Enterprise EAM Maturity

SR71 Consult recommends a structured 4-wave transformation roadmap to bridge the identified capability gaps. Each wave builds on the previous, ensuring that foundational governance is in place before advanced analytics and predictive capabilities are introduced. The roadmap is designed to deliver measurable business value at each milestone.

1

Foundation & Governance

MONTHS 1-4 | TARGET: 55/100

- ▶ Asset Criticality scoring model (Risk Matrix)
- ▶ Functional location hierarchy (ISO 14224 aligned)
- ▶ Failure code taxonomy (Class / Cause / Remedy)
- ▶ Mandatory WO close-out audit controls
- ▶ Actual vs. estimated labor & materials reconciliation
- ▶ Technical inspection checklist enforcement
- ▶ Asset replacement value tracking

2

Maintenance Excellence

MONTHS 5-9 | TARGET: 65/100

- ▶ Maintenance Planner Workbench
- ▶ Resource-constrained scheduling engine
- ▶ Artisan skills matrix & certification tracking
- ▶ Workforce capacity planning dashboard
- ▶ Schedule adherence KPI reporting
- ▶ Backlog age analysis and priority queuing
- ▶ Repairable spares management module

3

Reliability Excellence

MONTHS 10-15 | TARGET: 75/100

- ▶ FMEA library management and linkage to assets
- ▶ RCA investigation workflow (5-Why / Fishbone)
- ▶ Repeat failure detection and auto-escalation
- ▶ Reliability KPI dashboards (MTBF, MTTR, OEE)
- ▶ RCM decision logic integration to PM strategy
- ▶ Critical spares classification and stockout alerts
- ▶ Maintenance strategy review cycle automation

4

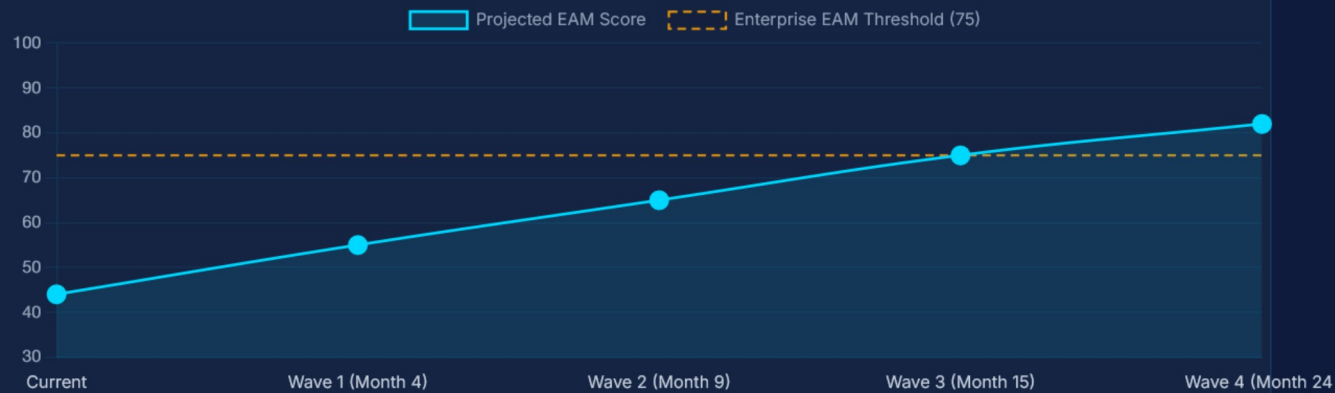
Asset Performance Management

MONTHS 16-24 | TARGET: 82/100

- ▶ IoT sensor data integration (Azure IoT Hub)
- ▶ Condition-Based Maintenance (CBM) trigger engine
- ▶ Predictive Maintenance (PdM) ML model integration
- ▶ Remaining Useful Life (RUL) estimation dashboard
- ▶ Digital twin asset health visualization
- ▶ ERP integration (SAP / Dynamics 365)
- ▶ ISO 55001 certification readiness assessment

Projected Maturity Progression

EAM Maturity Progression — 4-Wave Roadmap



WAVE	TIMELINE	PROJECTED SCORE	EAM MATURITY LEVEL	KEY BUSINESS VALUE
Current State	—	44 / 100	Level 2.2 — Transitioning	Strong CMMS for facilities management
Wave 1 Complete	Month 4	55 / 100	Level 3 — Planned Maintenance	Structured failure data; audited close-outs
Wave 2 Complete	Month 9	65 / 100	Level 4 — Maintenance Excellence	Resource optimization; schedule compliance
Wave 3 Complete	Month 15	75 / 100	Level 5 — Reliability Excellence	Measurable MTBF/MTTR improvement; RCA-driven
Wave 4 Complete	Month 24	82 / 100	Level 6 — Asset Performance Mgmt	Predictive maintenance; ISO 55001 ready

A mature EAM platform must support the measurement, trending, and governance of a comprehensive set of reliability and maintenance performance indicators. The following KPI framework defines the metrics that the DesireInfoWeb platform must support to meet enterprise EAM standards. Currently, the platform supports approximately 30% of these metrics through its existing reporting module.

Tier 1: Asset Reliability KPIs

MTBF

Mean Time Between Failures

Average operating time between unplanned failures. Primary indicator of asset reliability. Requires structured failure coding at WO close-out.

MTTR

Mean Time To Repair

Average time to restore an asset to operational status. Measures maintenance response and repair efficiency. Requires actual labor time capture.

OEE

Overall Equipment Effectiveness

Composite measure of Availability × Performance × Quality. The gold standard for manufacturing asset performance. Requires production data integration.

Asset Availability

Uptime / (Uptime + Downtime)

Percentage of time an asset is available for production. Directly linked to failure frequency and repair time. Critical for SLA management.

Failure Rate

Failures per Operating Hour

Frequency of unplanned failures over time. Used to identify deteriorating assets and validate maintenance strategy effectiveness.

RUL

Remaining Useful Life

Predicted remaining operational life of an asset before failure. Requires condition monitoring data and predictive analytics models.

Tier 2: Maintenance Performance KPIs

Schedule Adherence

Completed PM / Scheduled PM × 100

Percentage of planned maintenance tasks completed on schedule. Target: >95%. Currently not measurable in the DesireInfoWeb platform.

PM Compliance

PM WOs Completed on Time / Total PM WOs

Measures adherence to the preventive maintenance programme. Directly impacts asset reliability and warranty compliance.

Backlog Age

Average Age of Open Work Orders

Measures the accumulation of deferred maintenance. High backlog age indicates resource constraints or priority management failures.

Reactive Ratio

Reactive WOs / Total WOs × 100

Percentage of work that is unplanned/ reactive. World-class target: <20%. High reactive ratio indicates poor maintenance strategy.

Maintenance Cost / RAV

Annual Maint. Cost / Replacement Asset Value

Maintenance spend as a percentage of asset replacement value. World-class benchmark: 2–4%. Requires asset valuation data.

Repeat Failure Rate

Repeat Failures / Total Failures × 100

Percentage of failures that recur within a defined period. High rate indicates ineffective root cause elimination. Requires failure coding.



Shutdown & Turnaround (S&T) / Opportunity Maintenance

Project Planning Module Requirements & Capability Gap Analysis

Annual shutdowns, planned turnarounds, and opportunity maintenance windows represent the highest-risk, highest-cost events in any industrial asset management program. A single unplanned day of extended downtime during a shutdown can cost millions in lost production. This discipline demands a dedicated **Project Planning Module** — a capability category that is entirely absent from the DesireInfoWeb platform as currently proposed.

Critical Gap: The platform processes all maintenance as discrete, flat work orders. It provides no hierarchical project structure, no critical path scheduling, no resource leveling across thousands of concurrent tasks, no material kitting and staging, and no integration with industry-standard project scheduling tools (Primavera P6 / MS Project). These are non-negotiable requirements for any client operating in mining, oil & gas, manufacturing, utilities, or heavy industry.

S&T Capability Scoring

S&T PROJECT PLANNING

1/10

No project hierarchy, WBS, or CPM scheduling exists. All work is flat work orders.

OPPORTUNITY MAINTENANCE

2/10

No automated backlog grouping when assets go offline unplanned. Manual workaround only.

MATERIAL KITTING & STAGING

4/10

Basic inventory reservation exists. No "Bag and Tag" or pre-shutdown kitting workflow.

CONTRACTOR COORDINATION

5/10

Vendor portal supports task assignment. Lacks multi-contractor sequencing and gate management.

Detailed Capability Gap Analysis

11.1 Project Hierarchy & Work Breakdown Structure (WBS)	
OBSERVED IN PROPOSAL	CRITICAL GAP
Flat work order management. Work orders link to assets but not to parent projects. No concept of a "Turnaround Project" or "Shutdown Event" as a container entity.	A full WBS hierarchy is required: Shutdown Event → Discipline (Mechanical / Electrical / Instrumentation) → Work Package → Task. Thousands of work orders must roll up to a single project code for financial tracking, progress reporting, and executive dashboards.

11.2 Critical Path Method (CPM) Scheduling

OBSERVED IN PROPOSAL

Basic calendar view for individual work orders. Scheduled date and estimated duration fields exist at work order level.

CRITICAL GAP

Full Gantt chart visualization with task dependencies (Finish-to-Start, Start-to-Start, Finish-to-Finish, lag/lead time) is required. The system must calculate and highlight the critical path, flag schedule float, and support what-if scenario planning. Alternatively, a certified bi-directional API integration with Primavera P6 or MS Project is required.

11.3 Opportunity Maintenance Execution

OBSERVED IN PROPOSAL

Work orders can be created ad hoc. Backlog management exists at a basic level. No automated triggers on asset status changes.

CRITICAL GAP

When an asset goes offline unexpectedly, the system must automatically surface all "Opportunity Maintenance" backlog items tagged to that asset and its functional location. The planner must be able to generate an execution package with one click, including labour, materials, and permits — enabling the maintenance team to execute opportunistic work before the asset is returned to service.

11.4 Material Kitting, Staging & "Bag and Tag"

OBSERVED IN PROPOSAL

Inventory management with reservation capability. Parts can be linked to work orders. Basic stock-level management with reorder point triggers.

CRITICAL GAP

Shutdowns require material to be physically kitted and staged weeks before execution. The system must support "Bag and Tag" workflows: reserving specific serialised parts against specific work orders, generating picking lists by work package, tracking physical staging locations in the warehouse, and confirming kit completeness before the shutdown start date. Missing a critical part on Day 1 of a shutdown is a catastrophic failure.

11.5 Permit to Work (PTW) Integration

OBSERVED IN PROPOSAL

Work orders include basic approval workflows. No mention of safety permitting, isolation management, or lock-out/tag-out (LOTO) procedures.

CRITICAL GAP

All shutdown work requires formal Permit to Work issuance, energy isolation certificates (LOTO), confined space entry permits, and hot work permits. These must be digitally integrated with the work order, with the work order blocked from execution until all required permits are issued and active. PTW is a legal and safety-critical requirement in all regulated industries.

11.6 Multi-Contractor Gate & Progress Management

OBSERVED IN PROPOSAL

Vendor portal supports task assignment and completion confirmation. Proximity-based vendor dispatch for reactive work.

CRITICAL GAP

During a shutdown, 10–50 contractors work simultaneously in a sequenced, interdependent schedule. The system must support contractor gate management (daily headcount, induction status, access control), progress reporting by discipline and work package, earned value tracking, and automated escalation when a contractor falls behind the critical path. The current vendor portal is designed for reactive, single-vendor dispatch — not multi-contractor outage coordination.

Project Planning Module: Minimum Viable Specification

The following table defines the minimum required capabilities for a Project Planning Module that would make the DesireInfoWeb platform viable for clients with annual shutdown requirements.

CAPABILITY	PRIORITY	INDUSTRY STANDARD REFERENCE	BUILD VS. INTEGRATE
Shutdown / Turnaround Project entity with WBS	P1 — CRITICAL	SMRP Best Practice 2.2.4	Build native
Gantt chart with task dependencies & CPM	P1 — CRITICAL	PMI PMBOK / Primavera P6	Build or integrate P6/MS Project
Opportunity Maintenance backlog surfacing	P1 — CRITICAL	SMRP Best Practice 2.2.2	Build native
Material kitting, staging & Bag and Tag	P1 — CRITICAL	SMRP Best Practice 4.1.3	Build native (extend inventory module)
Permit to Work (PTW) & LOTO integration	P1 — CRITICAL	ISO 45001 / OSHA 1910.147	Build native or integrate specialist PTW system
Multi-contractor progress & gate management	P2 — HIGH	SMRP Best Practice 5.1.2	Extend vendor portal
Earned Value Management (EVM) reporting	P2 — HIGH	PMI PMBOK / ISO 21508	Build native (extend reporting module)
Shutdown cost vs. budget tracking	P2 — HIGH	ISO 55001 Clause 6.2	Build native (extend financial module)
Post-shutdown lessons learned register	P3 — MEDIUM	ISO 55001 Clause 10.1	Build native
Primavera P6 / MS Project API integration	P2 — HIGH	Industry standard	API integration (bi-directional)

SR71 Assessment: The absence of a Project Planning Module is the single largest barrier to DesireInfoWeb entering the heavy industry, mining, oil & gas, and manufacturing markets. Clients in these sectors will not adopt a CMMS that cannot manage their annual shutdown — the most critical and expensive maintenance event of the year. This module should be the **top priority in Wave 2** of the development roadmap.